Edge Intelligence in Healthcare

Revolutionizing Smart Healthcare with ML and Blockchain.

Mohammad Aazam, Ph.D.

Carnegie Mellon University.

aazam@ieee.org

maazam@cmu.edu

Intelligence vs Smartness

Intelligence

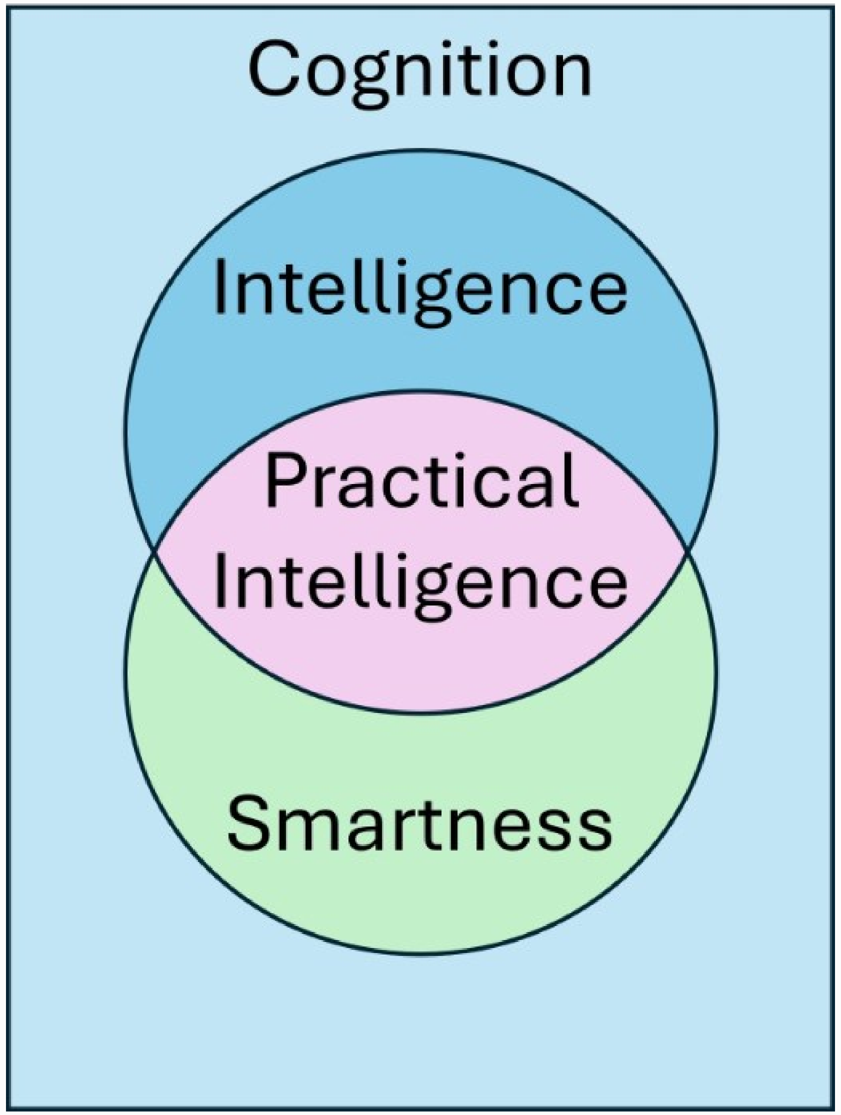
The innate or acquired ability to learn, understand, and apply knowledge or skills.

Associated with reasoning, problem-solving, and adaptability across various contexts.

Smartness

The ability to act or think quickly, often in specific or practical situations.

Closely tied to situational awareness and the ability to make sound judgments efficiently.



Edge Computing Overview

Decentralized Processing

Data processed closer to the source.

Reduced Latency

Faster response times for critical applications.

Cloud Integration

Complements centralized cloud infrastructure.

Importance of Edge Computing

Data Locality

Reduce latency for real-time analysis.

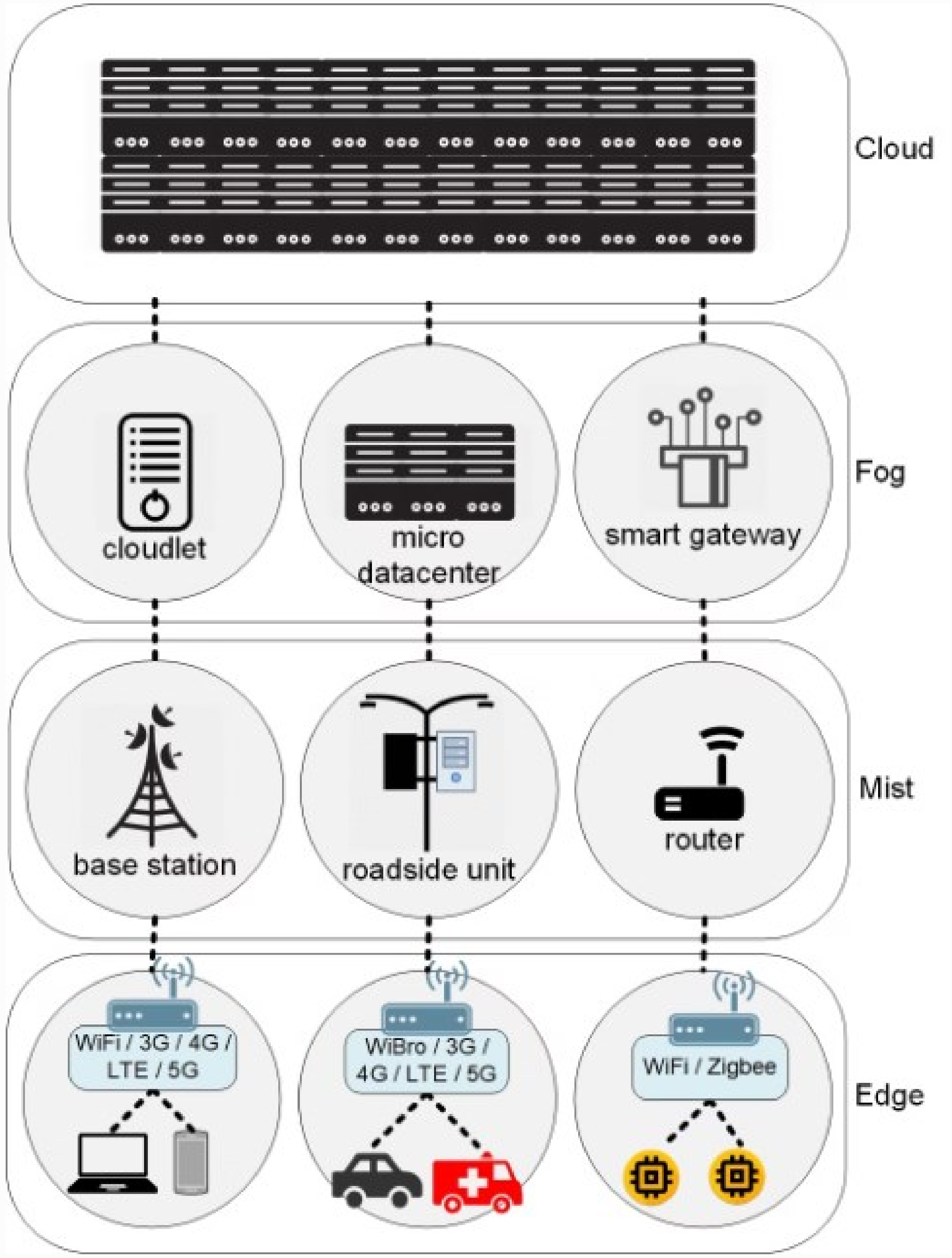
Improved Performance

Faster processing and quicker response times.

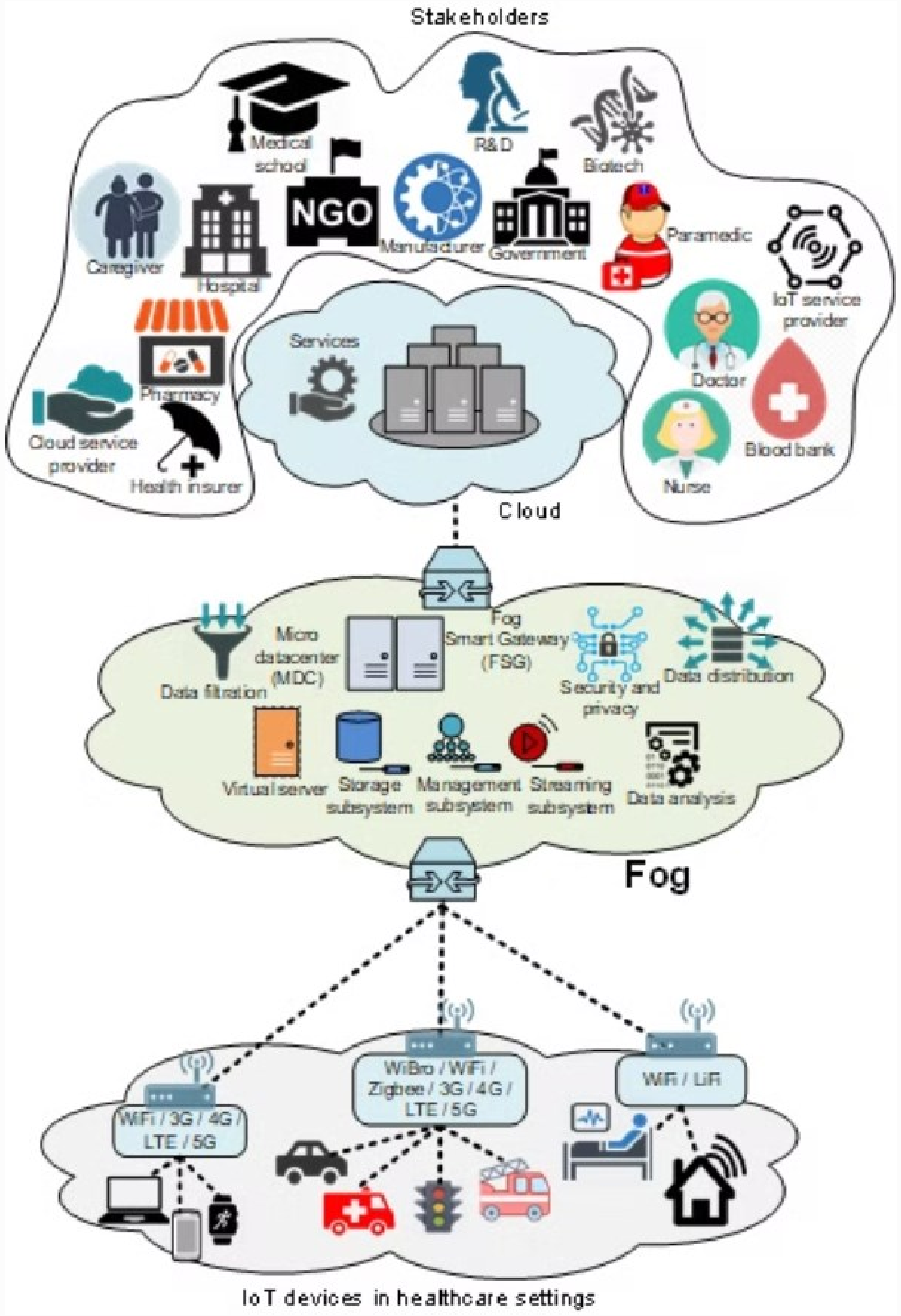
Enhanced Security

Minimize data transmission to external servers.

Cost optimization. More privacy-aware. More context-aware.

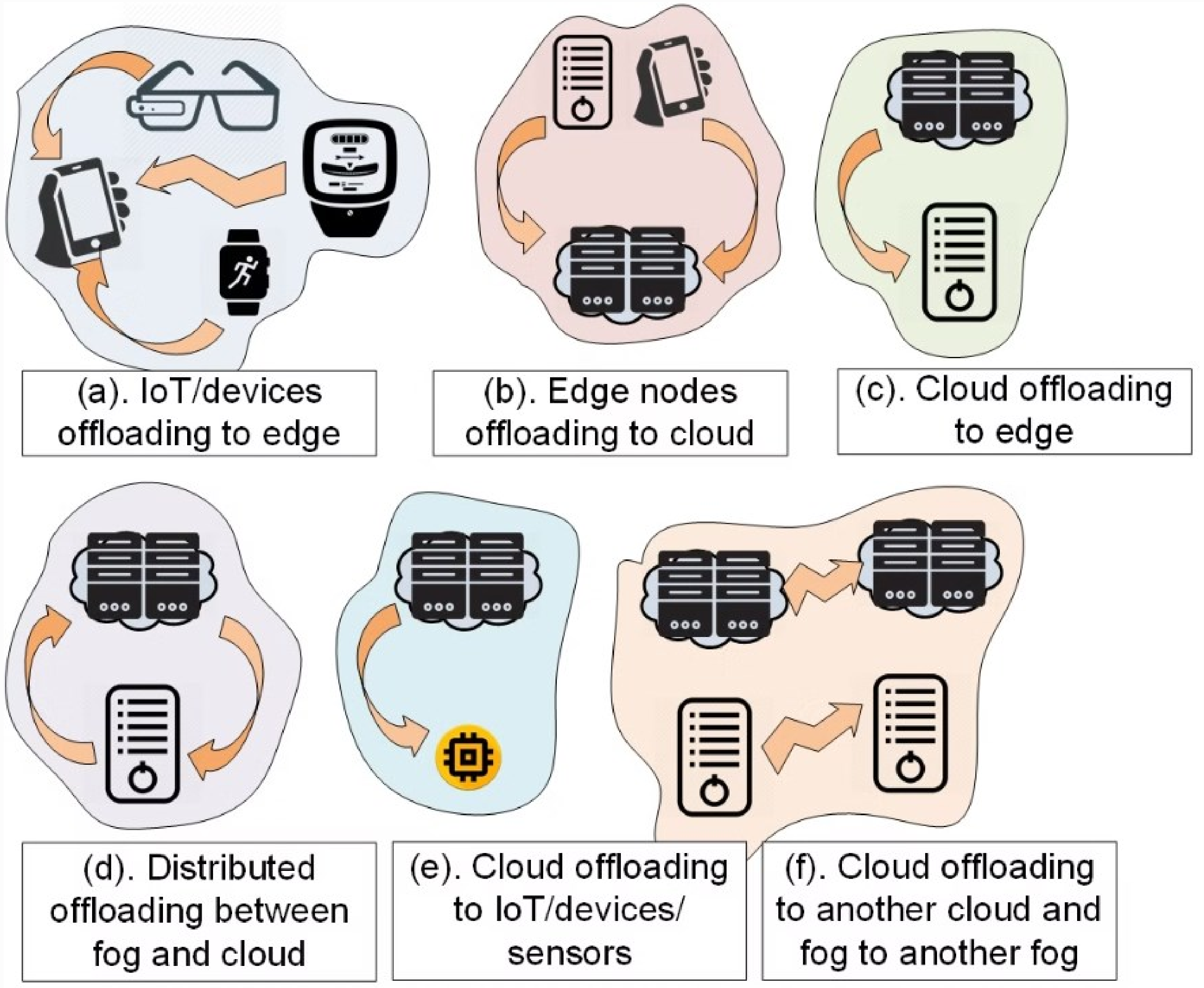


Aazam, M., Zeadally, S., & Harras, K. A. (2018). Offloading in fog computing for IoT: Review, enabling technologies, and research opportunities. Future Generation Computer Systems



Aazam, M., Zeadally, S., & Harras, K. A. (2020). Health Fog for Smart Healthcare. IEEE Consumer Electronics

Offloading Vectors



Resource Tradeoffs and Offloading

Edge Device Resources

Faster response times and enhanced privacy through local processing, but face resource constraints that can impact device performance when handling complex tasks.

Cloud Computing Resources

Excels at handling computationally intensive tasks and processing large datasets, though with potential tradeoffs in latency and data privacy.

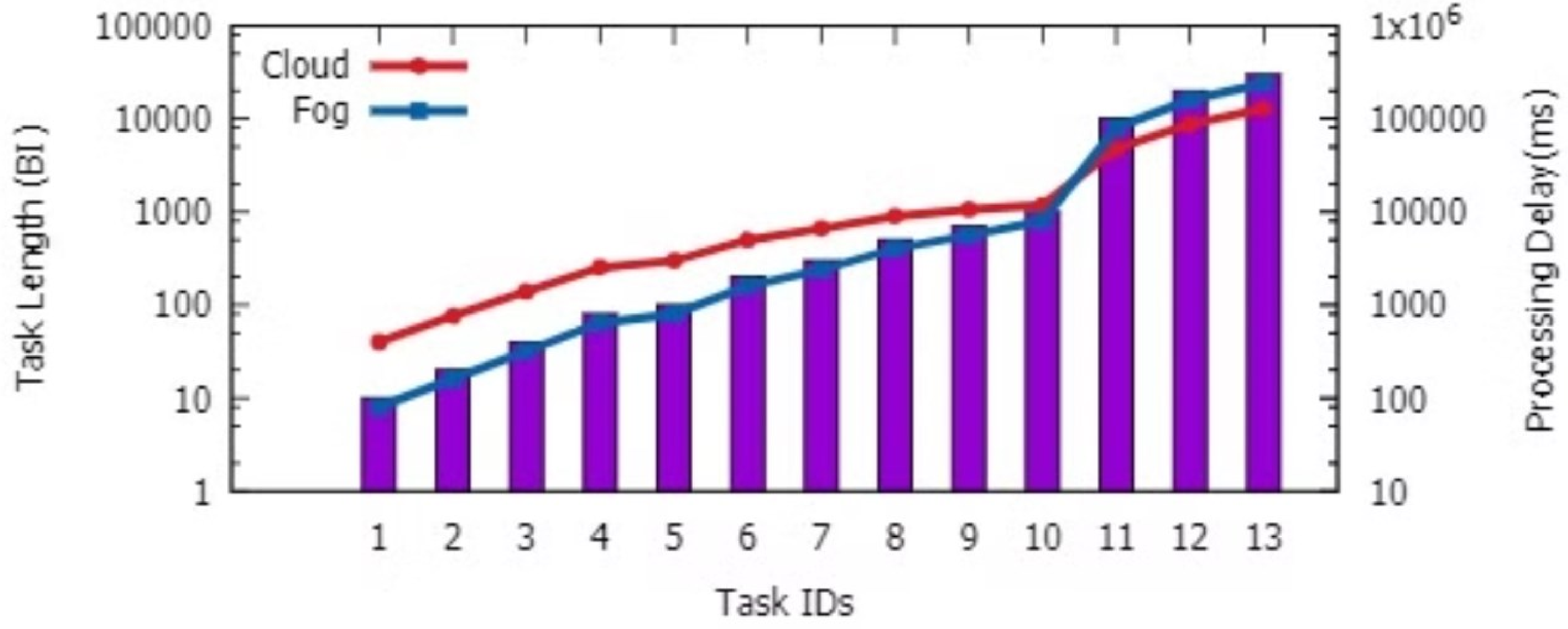


Figure 8: Tradeoff between computational reliability of fog vs cloud and the incurred delay.

Aazam, M., Harras, K. A., & Elgazar, A. E. (2018). Delay Tolerant Computing: The Untapped Potential. Proceedings of the 13th Workshop on Challenged Networks (ACM CHANTS)

Smart Healthcare: A Modern Approach

Personalized Care

Tailored to individual needs and preferences.

Data-Driven Decisions

Utilizes real-time information for better outcomes.

Integrated Systems

Connecting various healthcare technologies seamlessly.

Proactive Health Management

Focuses on prevention and early detection.

Introduction to Edge Intelligence

1 **Defining Edge Intelligence**

Combines edge computing and AI.

2 **Processing Power**

Data is processed closer to the source.

3 **Real-Time Insights**

Enables faster and more efficient decision-making.

4 **Improved Efficiency**

Reduces latency and improves overall system performance.

Edge vs Cloud vs Femtocloud

Cloud Computing

Centralized servers, high latency.

Femtocloud

Small-scale cloud setups, closer to the edge.

Edge Computing

Localized processing, reduced latency.

Examples of Edge Computing

Cloud

Radiology image storage for large-scale training.

Femtocloud

Opportunistic healthcare in remote areas for storing and processing data temporarily.

Edge

Real-time ECG analysis on a wearable device.

The Need for Edge Intelligence in Healthcare

- Latency caused by cloud dependency.
- Privacy concerns with centralized data storage.
- Connectivity issues in rural or underserved areas.

Healthcare Data Breaches: A Growing Threat

Healthcare data breaches have become a prevalent issue, posing significant risks to patient privacy and data security.

5,887

Breaches

Reported in 2023.

239%

Hacking Increase

From 2018 to 2023.

278%

Ransomware Increase

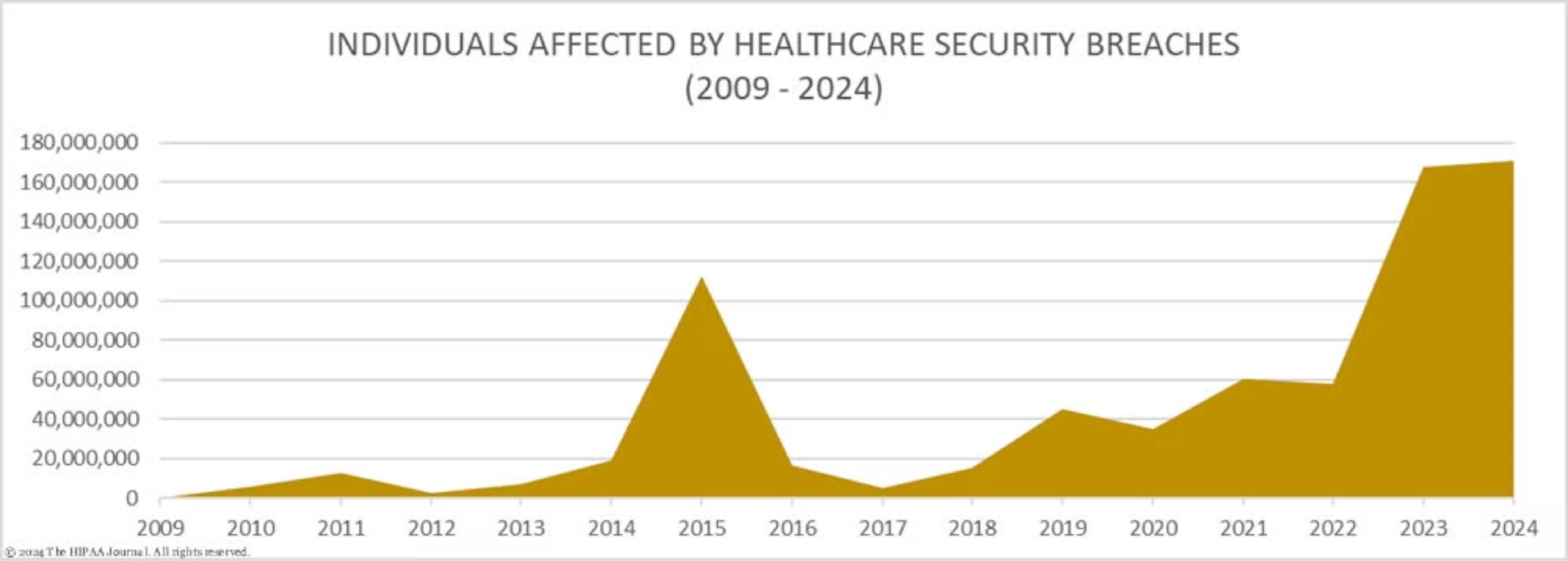
From 2018 to 2023.

79.7%

Hacking Breaches

In 2023.

HIPAA Journal, Nov. 2024.



HIPAA Journal, Nov. 2024.

Privacy Standards in Healthcare

1. **HIPAA**

   Protects sensitive patient health information.

2. **GDPR**

   EU regulation on data protection and privacy.

3. **CCPA**

   California's comprehensive privacy law.

4. **PIPEDA**

   Canada's law for personal information protection.

Research Directions

Blockchain Security

Secure data transmission using IoT sensors and blockchain.

Opportunistic Healthcare

Underserved areas using edge computing and femtocloud.

Data Privacy

How to maintain patient privacy while leveraging edge intelligence.

Opportunistic Healthcare (oHealth)

Opportunistic Healthcare

Opportunity-based. Usually, delay-tolerant.

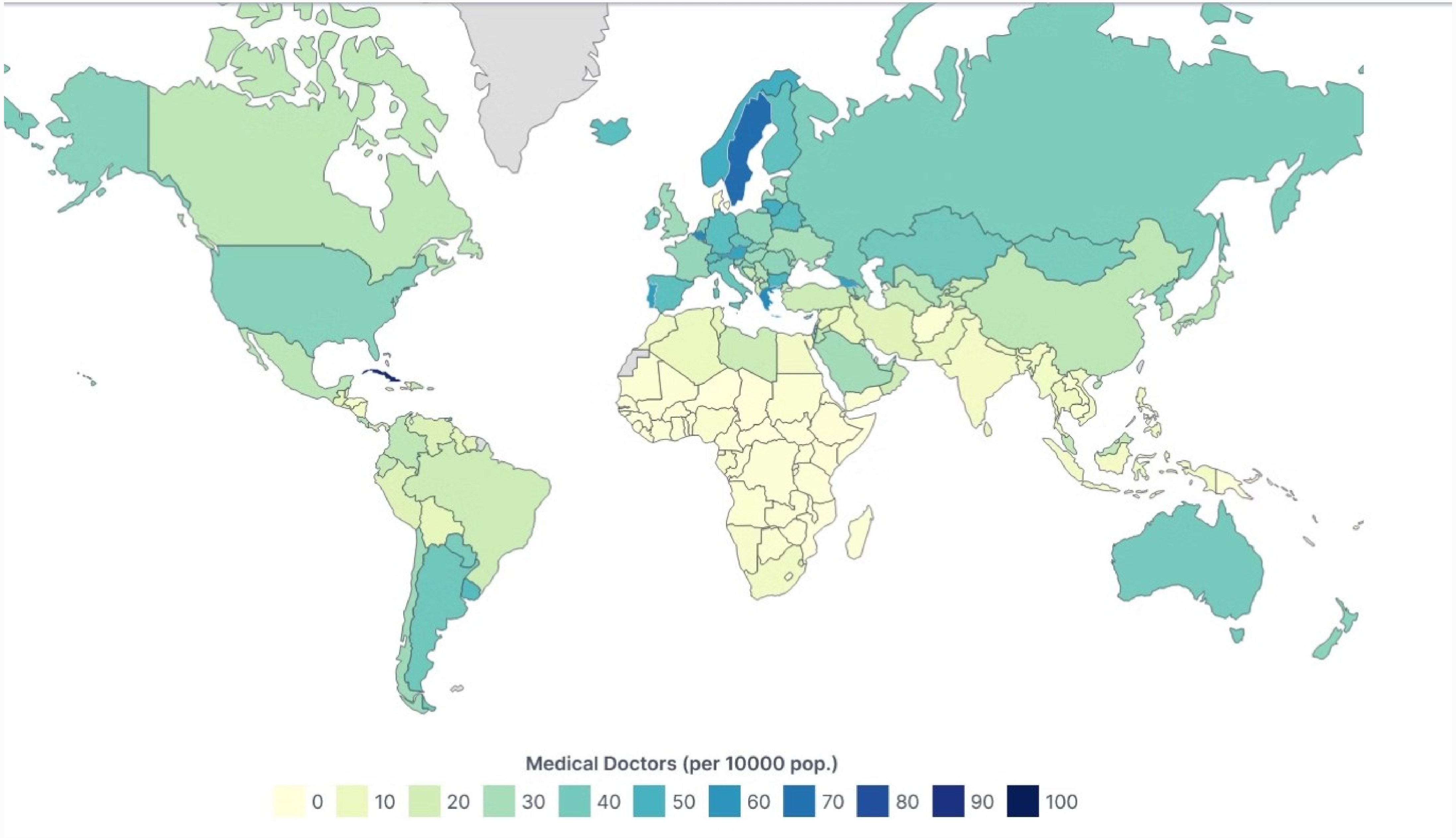
Edge / Fog Computing

Intermediate layer between edge and cloud.
Process data closer to the source.


Argentina
Travelling doctor takes healthcare to Argentina's remote areas
ALJAZEERA

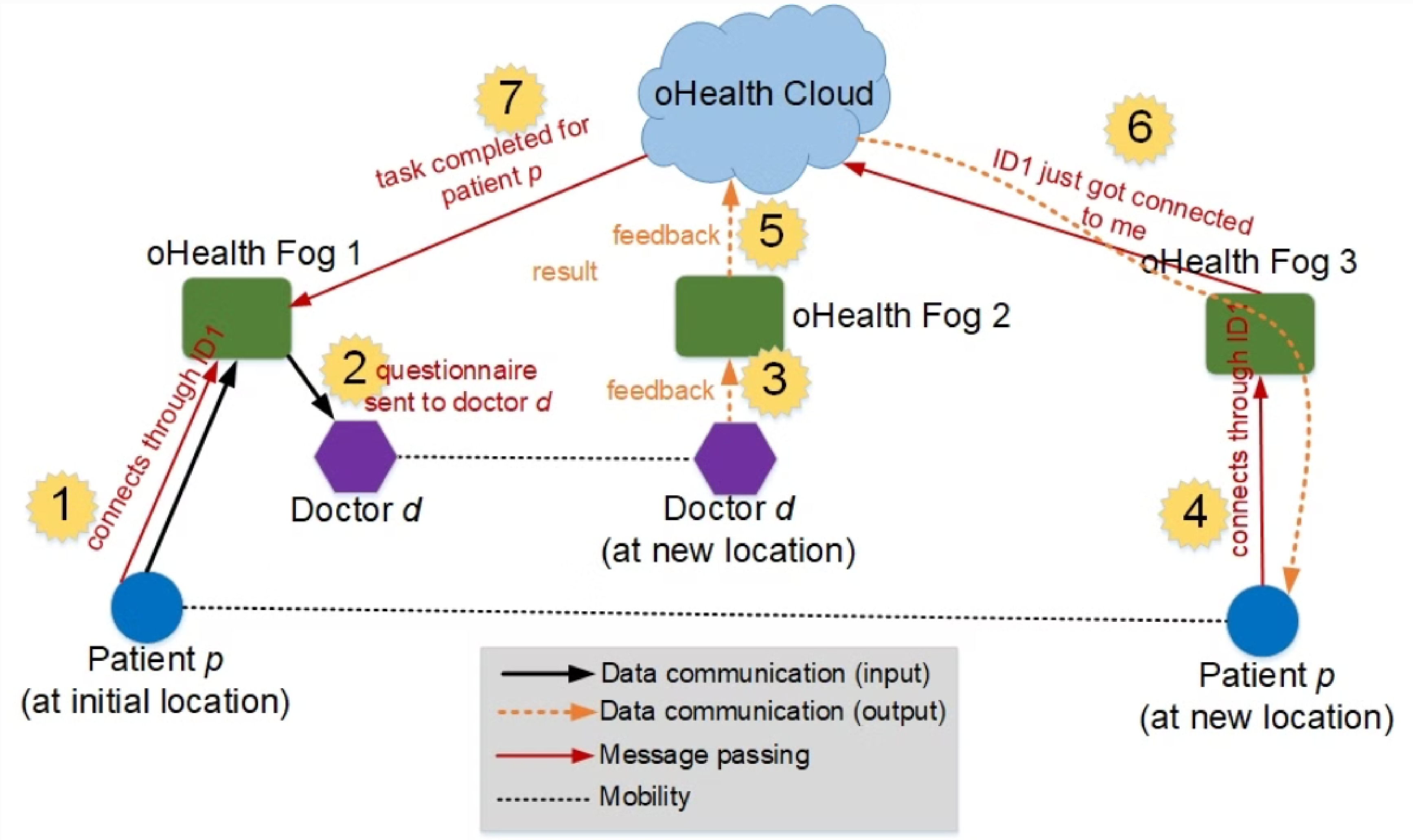

ALJAZEERA


ALJAZEERA

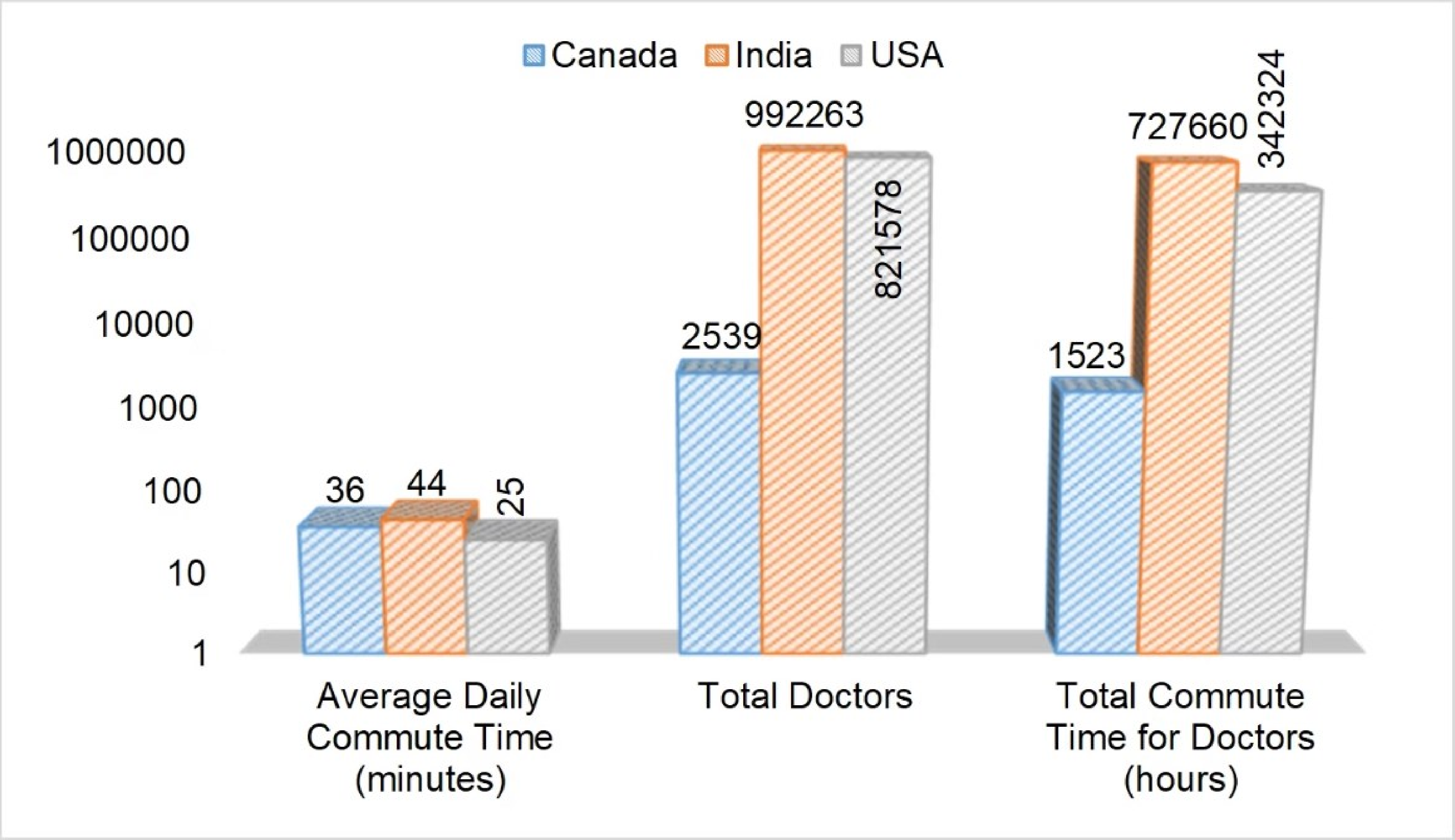


World Population Review 2024

oHealth



Aazam, M., & Fernando, X. (2019). oHealth: Opportunistic Healthcare in Public Transit through Fog and Edge Computing. Proceedings of the 4th IEEE International Conference on Smart Cloud (SmartCloud).



13,08,009 Doctors in India in 2024.

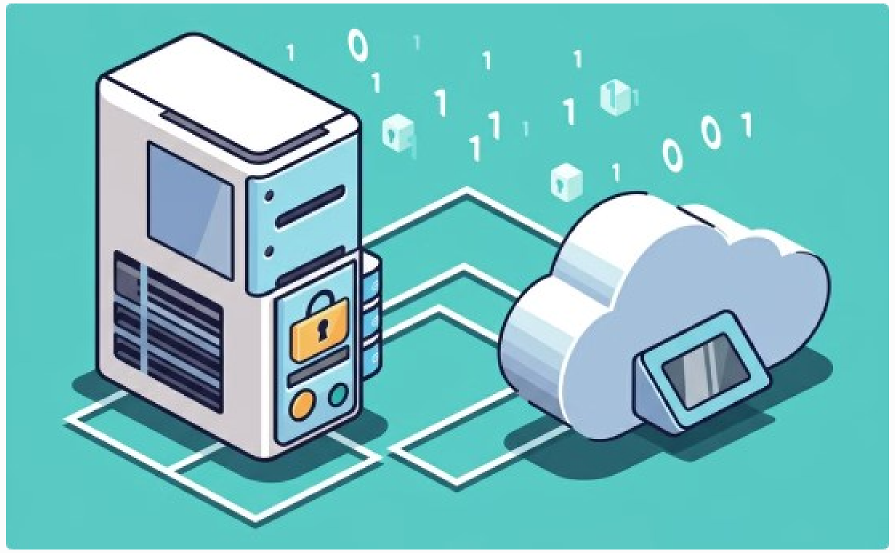
Delay-Tolerant Computing (DTC): Untapped Potential

Network Resilience

Data transfer despite interruptions

Mobile Connectivity

Ensures reliable data transfer



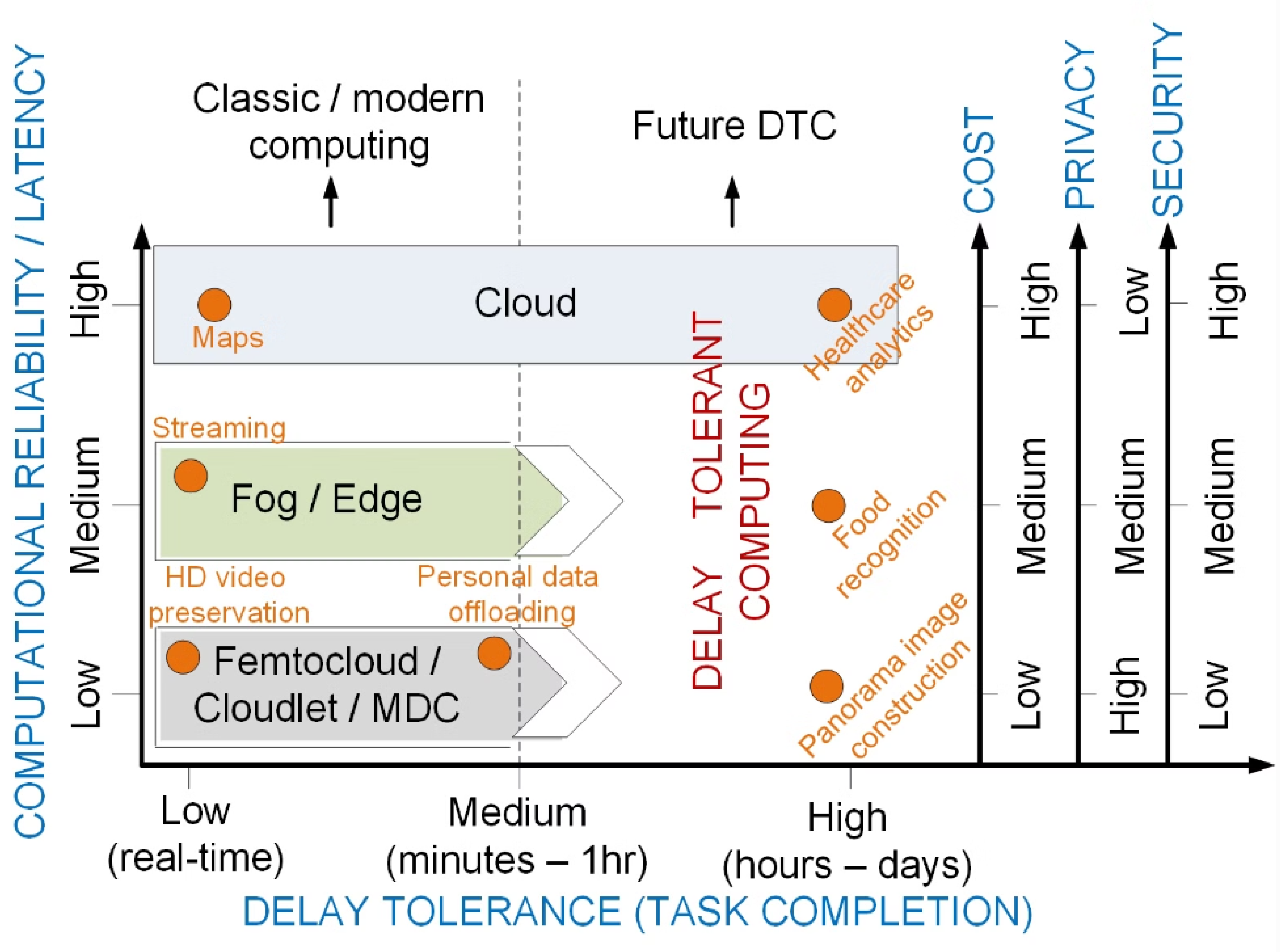
Healthcare Applications

Improve data transmission in challenging environments

Aazam, M., Harras, K. A., & Elgazar, A. E. (2018). Delay Tolerant Computing: The Untapped Potential. Proceedings of the 13th Workshop on Challenged Networks (ACM CHANTS)


UNDGONECTION
IOT
DT
IOT
IOT
IOT
IOT

From Classic Computing to DTC



Further Research Directions

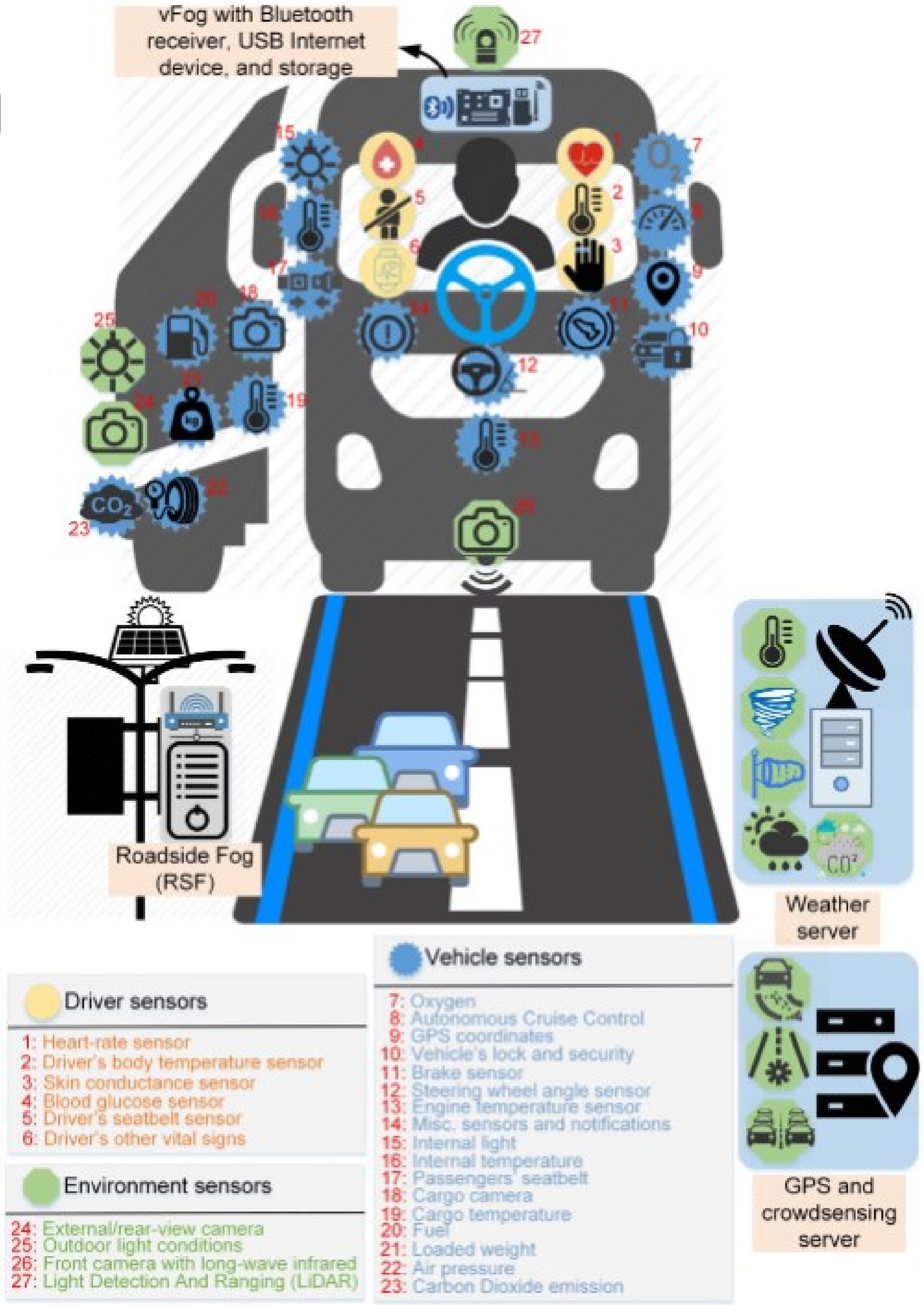
Energy-Efficient Femtocloud Architectures

Optimize cloud computing resources for remote healthcare.

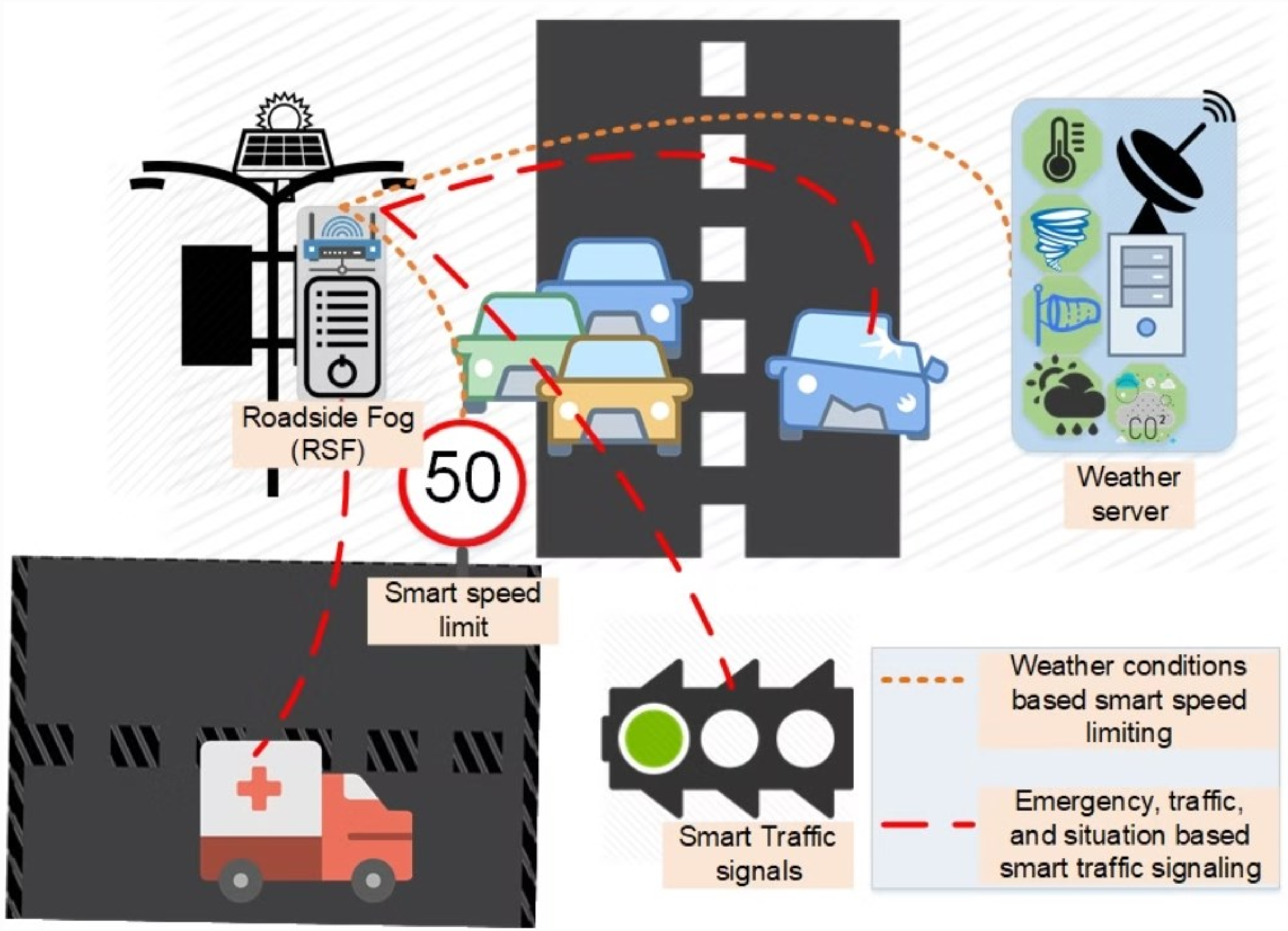
Opportunistic Healthcare

Leverage IoT devices for data collection and federated edge intelligence.

Driver Assistance, Monitoring, and Safety

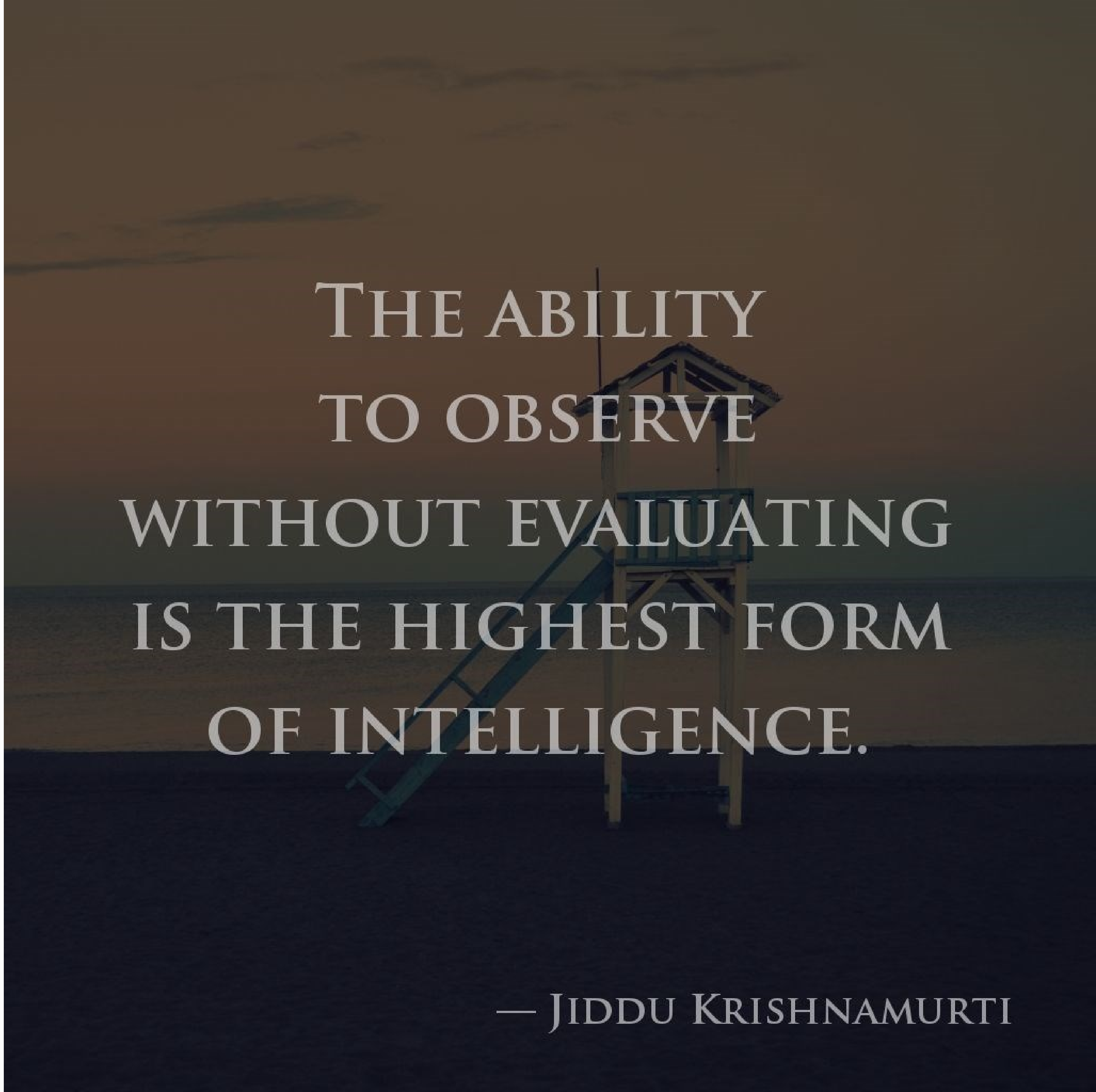


Aazam, M., & Fernando, X. (2017, September). Fog assisted driver behavior monitoring for intelligent transportation system. In 2017 IEEE 86th Vehicular Technology Conference (VTC-Fall)



Roadside Fog (RSF)
50
Smart speed limit
Weather server
Smart Traffic signals
Weather conditions based smart speed limiting
Emergency, traffic, and situation based smart traffic signaling

Quote Break


The ability
to observe
without evaluating
is the highest form
of intelligence.
— Jiddu Krishnamurti

Improving Patient Outcomes

Faster Diagnosis

Accurate results, reduced time.

Personalized Treatment

Tailored to patient needs, better outcomes.

Proactive Care

Detect risks early, prevent complications.

Reduced Hospital Stays

Faster recovery, lower costs.

Operational Efficiency

1

Resource Optimization

Improved resource allocation and use.

2

Streamlined Processes

Faster diagnosis and treatment times.

3

Cost Reduction

Lower operational expenses and patient costs.

Ensuring Data Privacy and Security

Data Protection

Sensitive medical records require robust security measures.

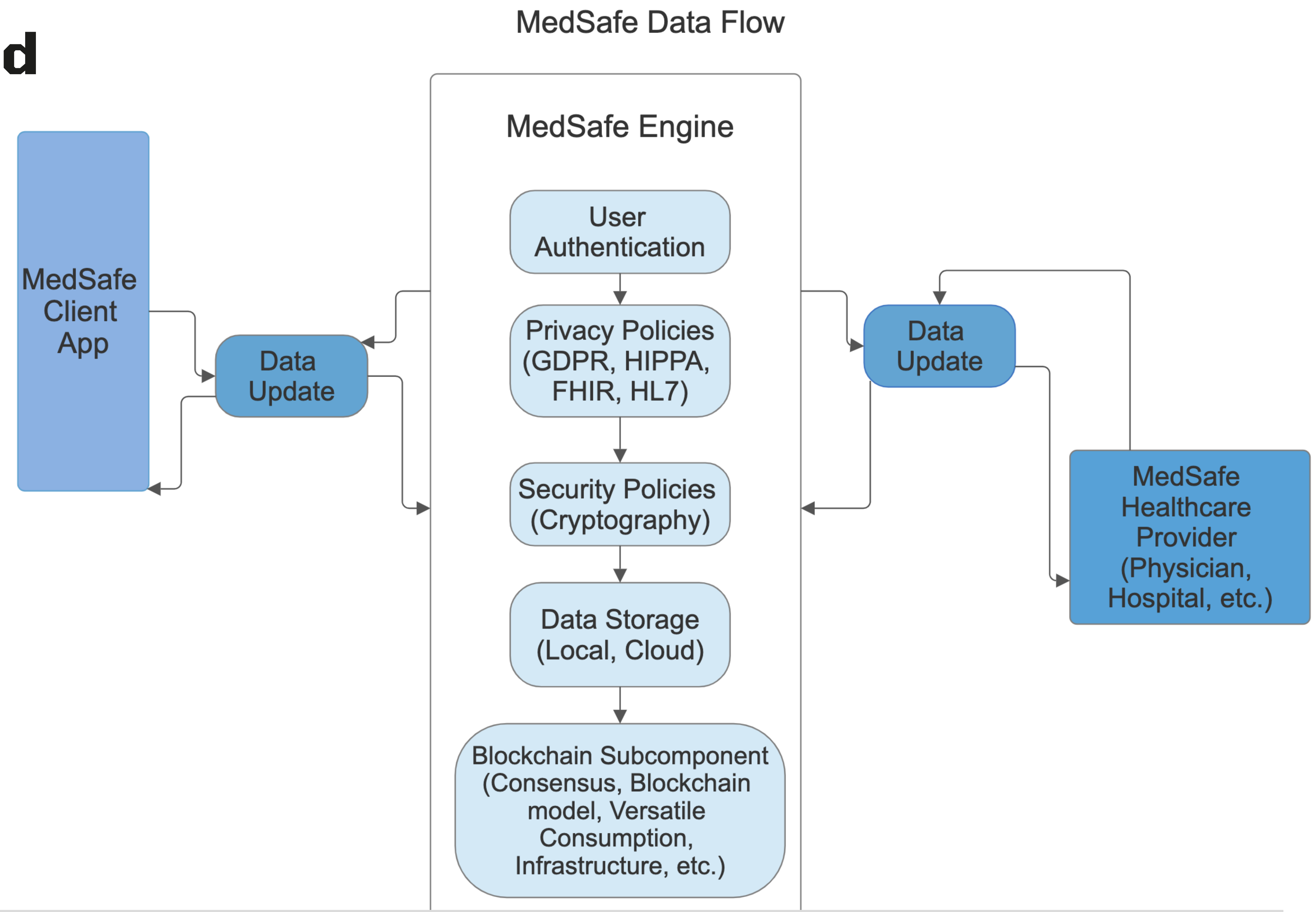
Privacy Compliance

Meeting HIPAA standards for data storage and transmission.

Secure Infrastructure

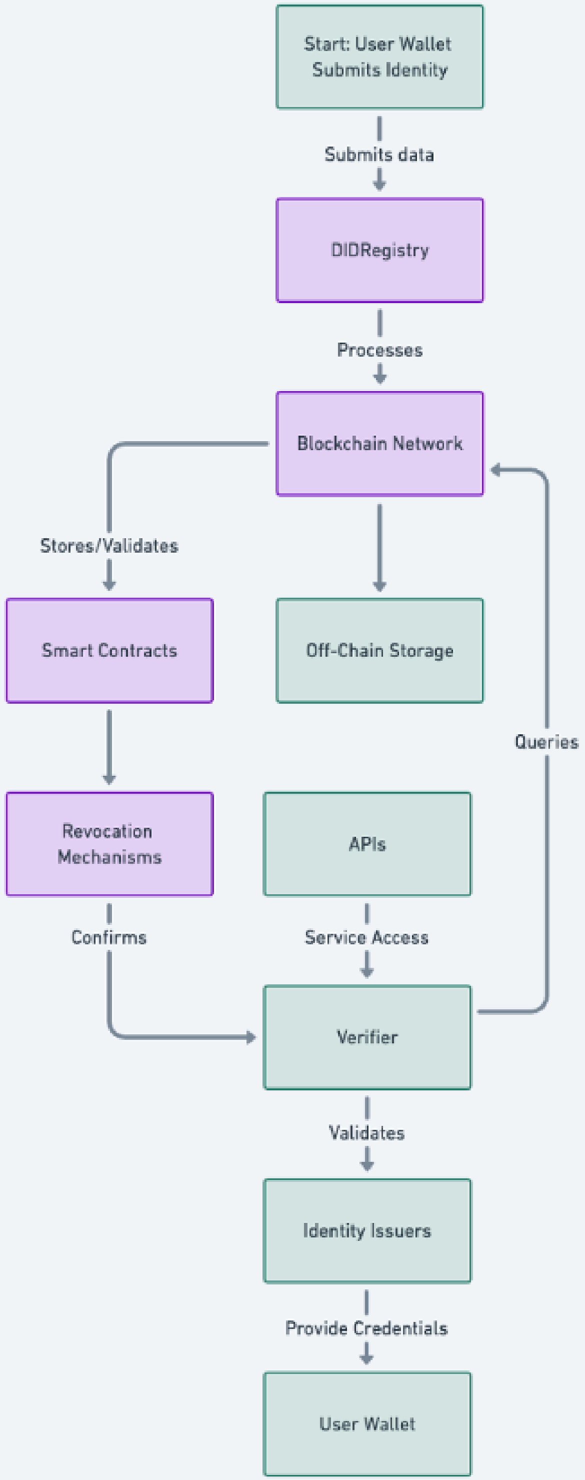
Protecting against unauthorized access and cyberattacks.

Data Privacy and Security Through Blockchain

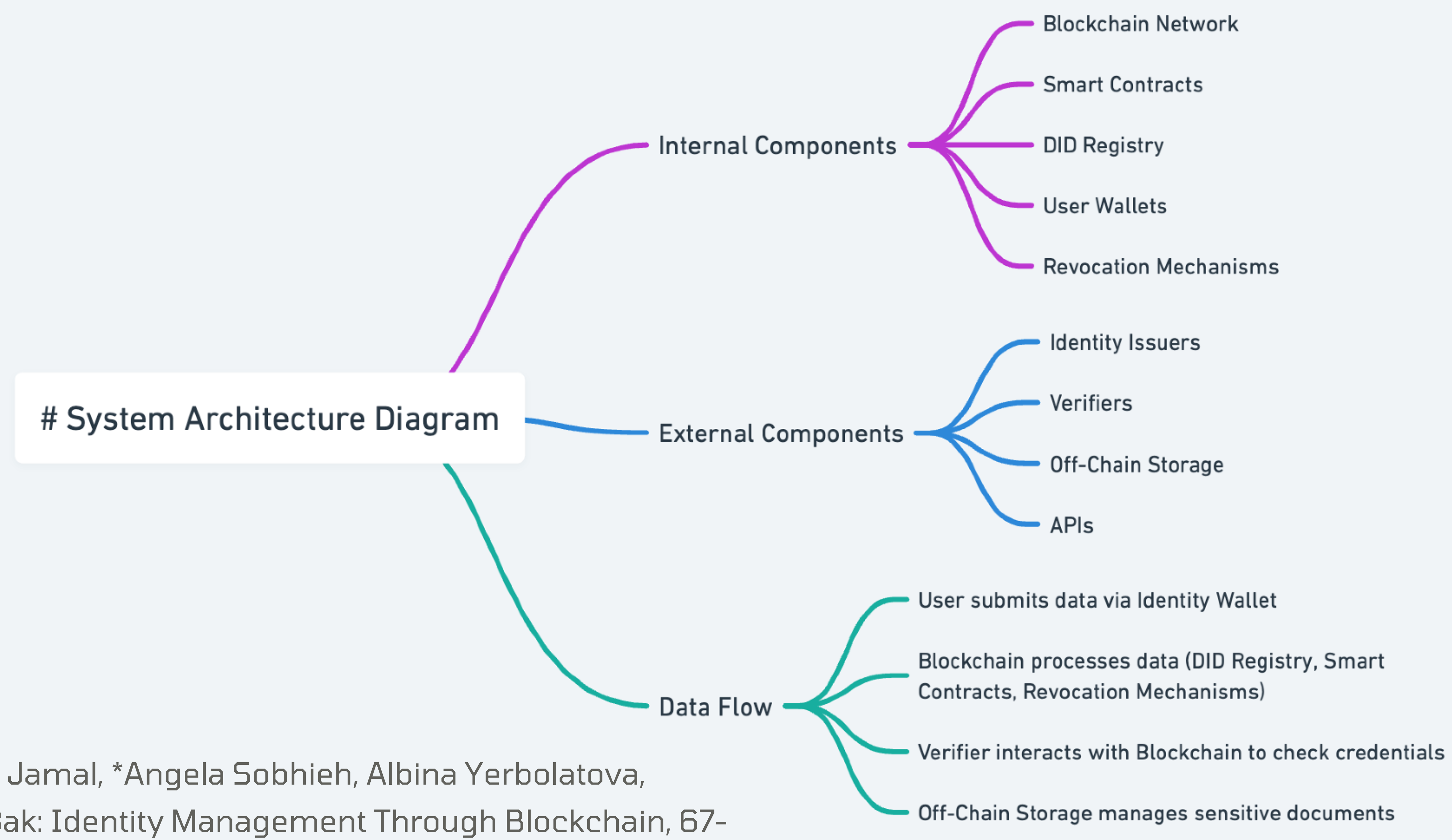


Laiba Sameer, Mohammad Aazam, "MedSafe: Blockchain-Driven Security and Traceability in Electronic Health Records", Summer Undergrad Research Apprenticeship, CMU, 2024.

Flexible Identity Management Through Blockchain



*Mona Jamal, *Angela Sobhieh, Albina Yerbolatova, IDMa3ak: Identity Management Through Blockchain, 67-404, Semester Project, CMU, Nov. 2024.
* Dual affiliation with American University of Beirut (AUB).



System Architecture Diagram
Internal Components
Blockchain Network
Smart Contracts
DID Registry
User Wallets
Revocation Mechanisms
External Components
Identity Issuers
Verifiers
Off-Chain Storage
APIs
Data Flow
User submits data via Identity Wallet
Blockchain processes data (DID Registry, Smart Contracts, Revocation Mechanisms)
Verifier interacts with Blockchain to check credentials
Off-Chain Storage manages sensitive documents

Components of Edge Intelligence

Devices

Smartwatches, glucose monitors, ECG monitors.

Edge Hardware

Raspberry Pi, NVIDIA Jetson Nano.

Communication

Bluetooth for wearables, LoRaWAN for rural healthcare.

Research Directions for Components

1

Lightweight CNNs

Optimization for wearable devices.

2

Blockchain Security

Secure communication for healthcare IoT.

3

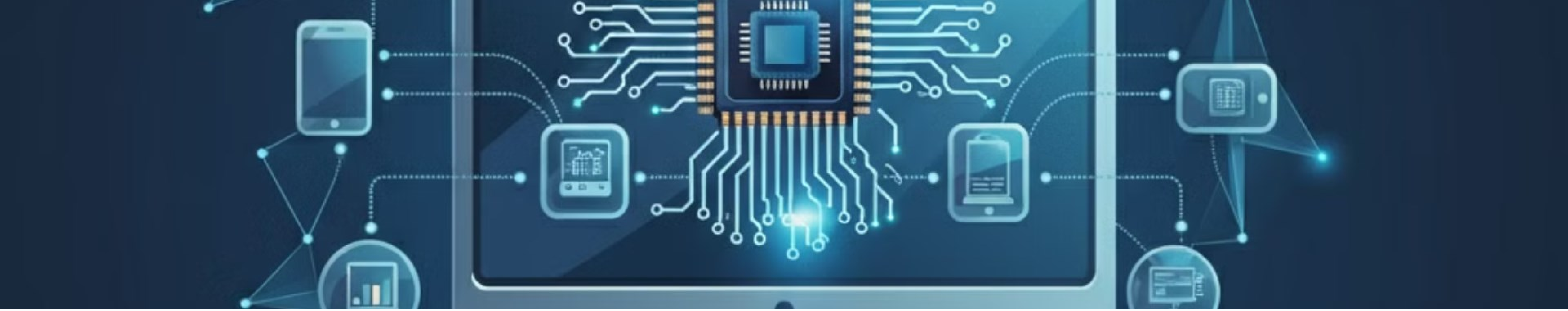
Efficient Processing

Raspberry Pi and Jetson devices.

4

Healthcare Wearables

Improve data analysis capabilities.



How ML Enhances Edge Computing

Lightweight ML Techniques

Reduce model size and complexity.

- Pruning
- Quantization

Resource-Constrained Devices

Enable deep learning on edge devices.

Improved Efficiency

Faster inference and reduced latency.

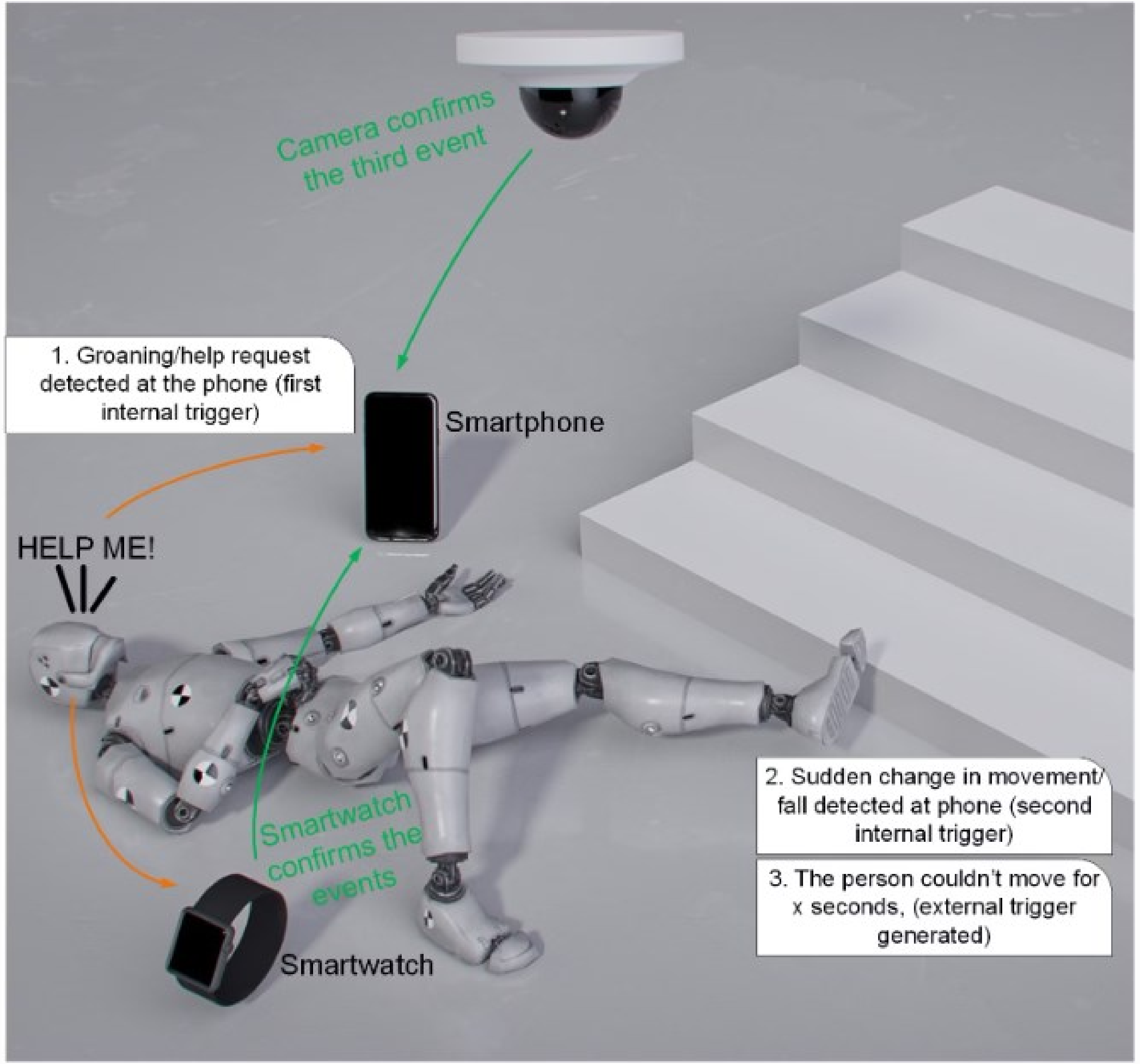
Applications of ML in Edge Computing

Fall Detection

ML models on wearables detect falls in elderly care.

Seizure Prediction

RNNs optimized for edge devices predict seizures.



Aazam, M., Zeadally, S., & Flushing, E. F. (2021). Task Offloading in Edge Computing for Machine Learning-based Smart Healthcare. Computer Networks

Further Research Directions in ML

Federated Learning

Train ML models across distributed edge devices.

Model Compression

Improve accuracy for edge applications.

Real-World Applications

Remote Patient Monitoring

Chronic disease management.

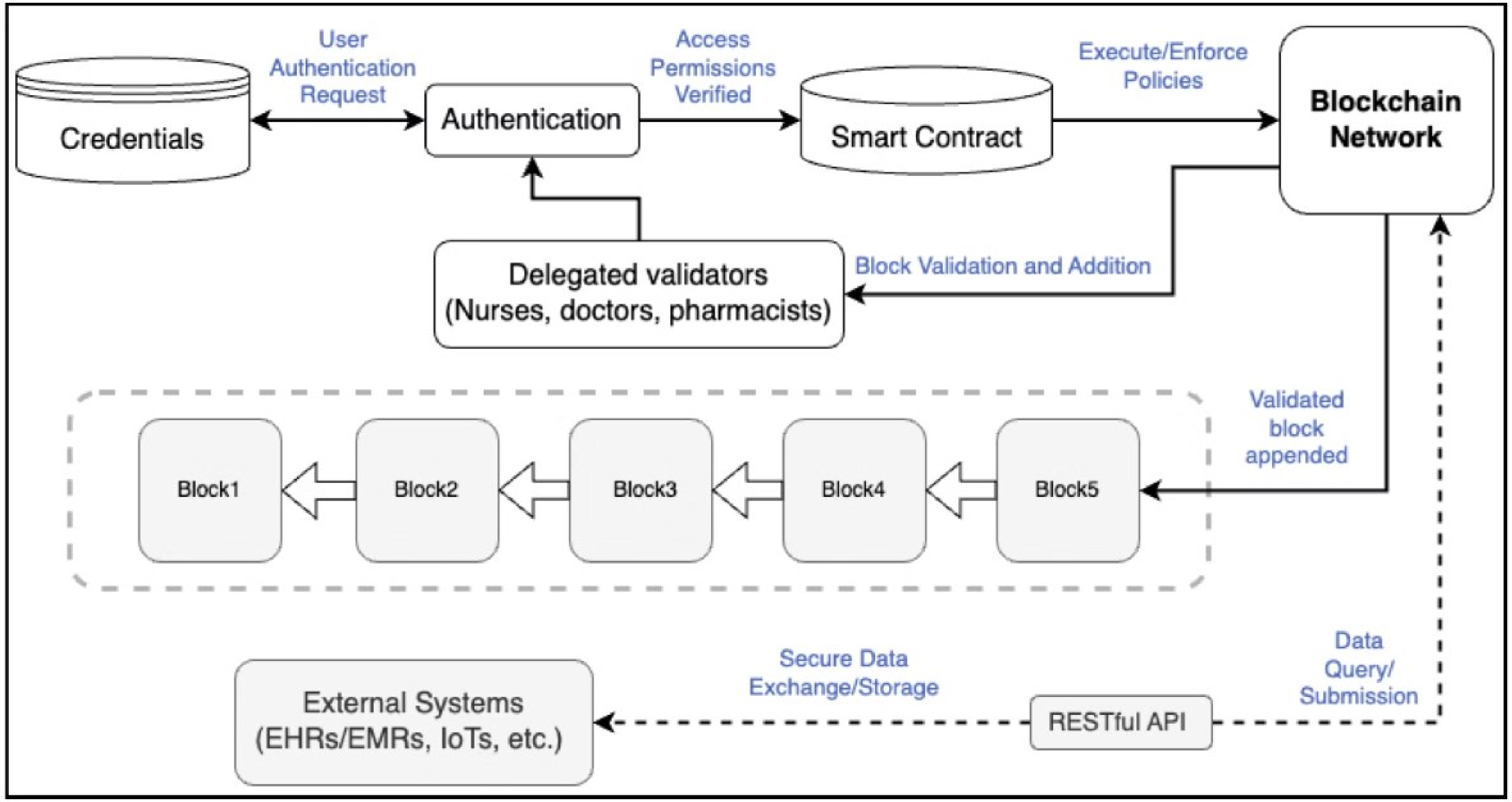
Predictive Diagnostics

Seizure and cardiac event prediction.

Blockchain-enabled EHRs

Secure, immutable patient records.

Blockchain for Smart Healthcare



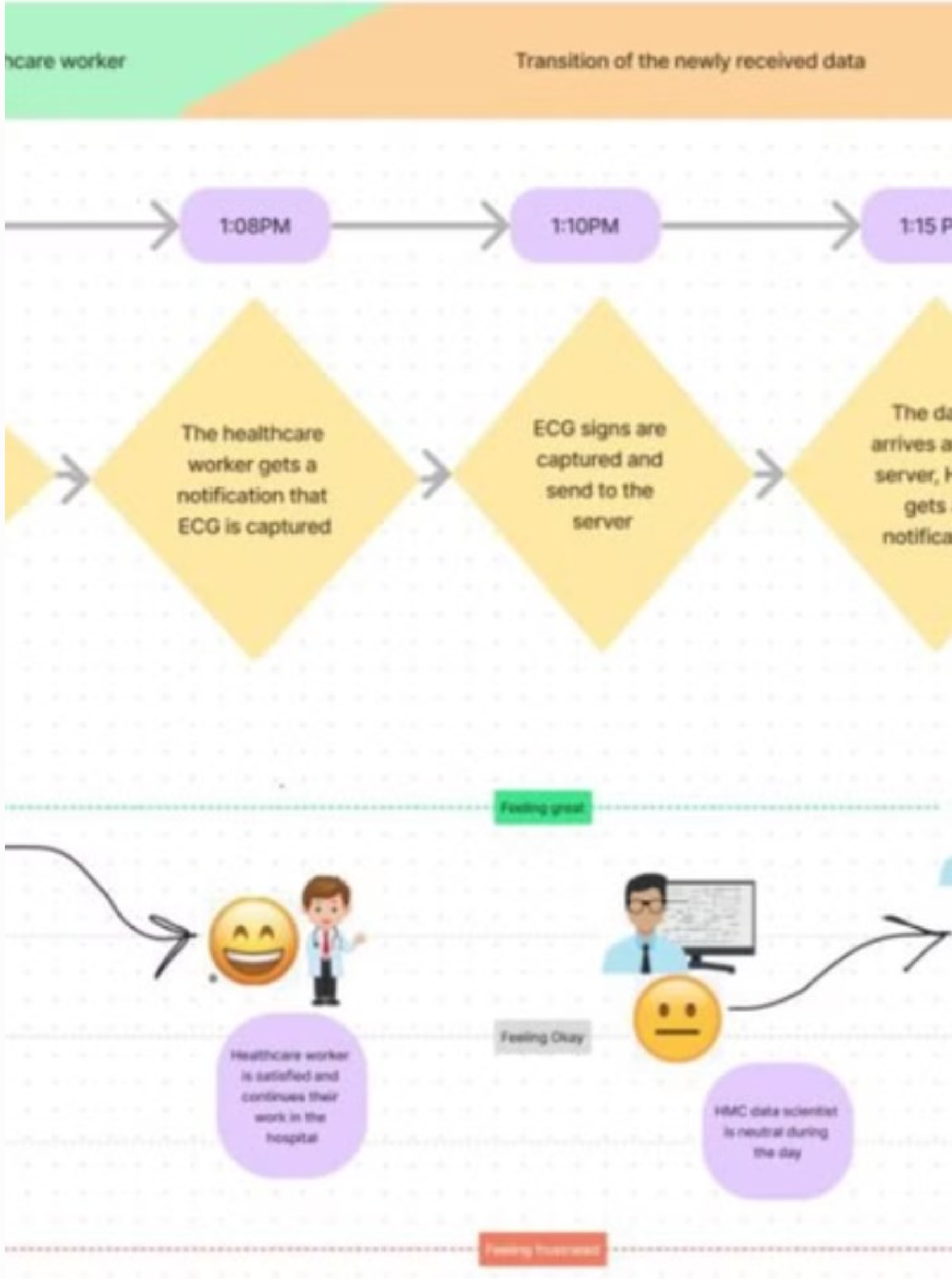
Aisha Al Attiyah, Latifa Al Hitmi, and Moza Al Thani, AafiyahChain: A Decentralized Healthcare Blockchain, 67-404, Semester Project, CMU, Nov. 2024.

Medical Service Providers' Healthcare

Healthcare providers suffer stress and burnout.

Generally, burnout scores ranged from 16% to 86%, with an overall mean burnout score of 57.4%. [Nature, May 2024.]

Muneera Al-Baker, Anastasia Cheypesh, Ahmad Mohamad,
TASMU Robotics, 67-373 Semester Project, CMU, March 2023.



Research Directions for Real-World Applications

Decentralized Data Sharing

Improve scalability of RPM systems.

Blockchain-based data sharing.

Opportunistic Platforms

Integrate blockchain and femtocloud.

Improved healthcare data management.

Case Studies in Smart Healthcare

Diabetes Monitoring

Edge ML models for glucose anomaly detection.

Cardiac Care

Detecting arrhythmia with Jetson Nano.

Research Directions for Case Studies

Blockchain Federated Learning

Cardiac monitoring across hospitals

Preserves patient privacy.

Opportunistic Healthcare Systems

Cardiac care for remote regions.

Improved access for underserved populations.

Advanced Imaging and Diagnostics

Edge Computing Power

Powerful imaging models run on affordable edge devices.

Real-Time Analysis

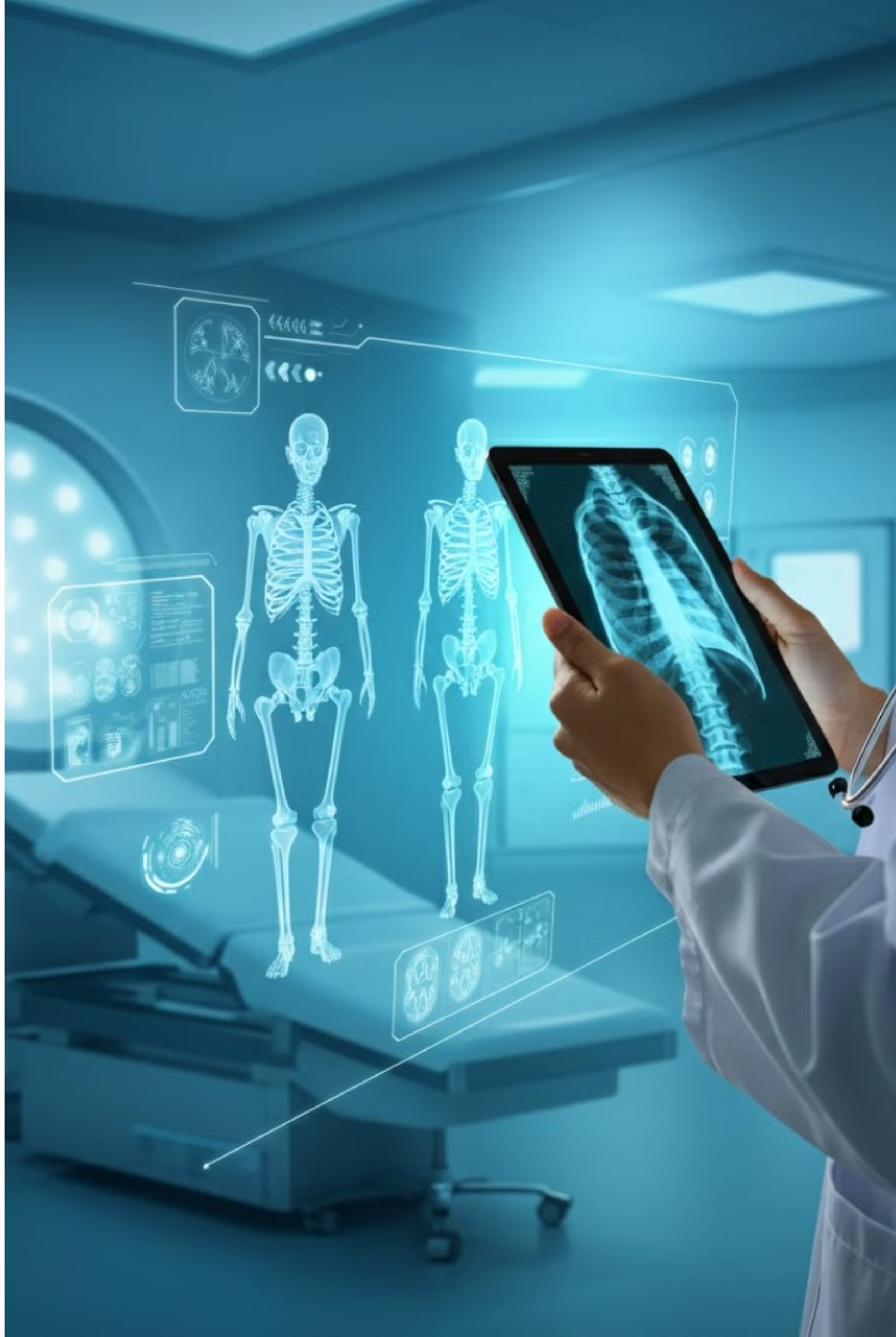
Enable rapid diagnosis in emergency settings.

MobileNetV2 Optimization

Efficient AI model for edge device processing.

Early Detection of Diseases

Enhance patient outcomes with timely interventions.



Research Directions for Imaging

AI-Powered Imaging

Improving diagnostic accuracy at the edge.

Femtocloud Integration

Short-term storage for rapid access.

Underserved Regions

AI-driven imaging diagnostics for remote areas.

Emergency Care and Mobility

Ambulance Triage

Machine learning in ambulances.

Real-time assessment of patient status.

Wearables for Elderly

Fall detection systems for home care.

Monitor vital signs and provide alerts.

Research Directions

Emergency Transport

Connect IoT and edge intelligence for continuous monitoring.

Secure Data Tracking

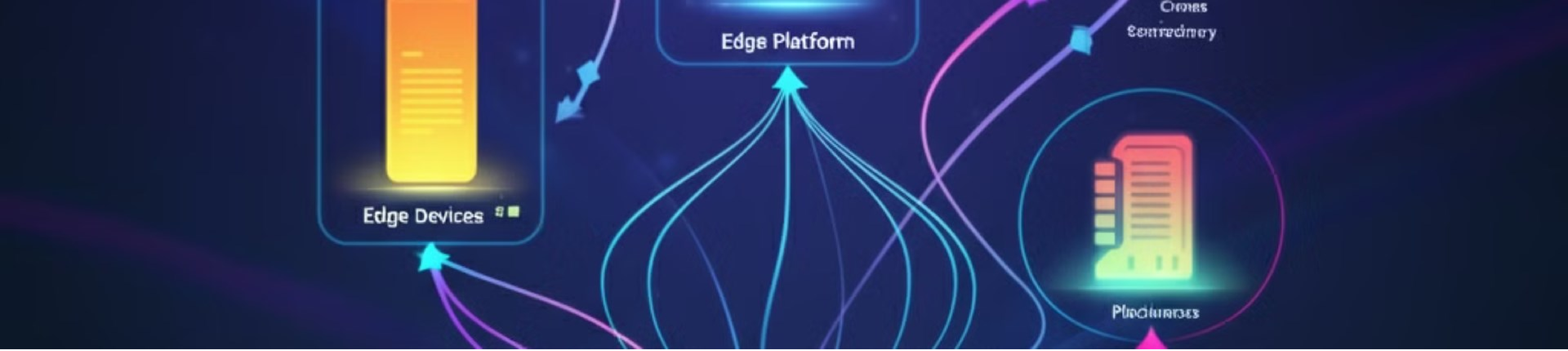
Enable blockchain for secure data storage and transfer.

Tools & Technologies

Edge devices and platforms.

ML frameworks.

Communication networks.



Edge Devices and Platforms

NVIDIA Jetson Nano

Real-time computer vision and NLP.

Google Coral

Supports TensorFlow Lite models for inference.

Edge TPU

Low-power ML inference on IoT devices.

Research Directions for Edge Devices

Blockchain-Enabled Architecture

Distributed computing for healthcare.

AI-Powered Predictive Analytics

Emergency healthcare with Edge TPU.

ML Frameworks for Edge

TensorFlow Lite

Lightweight models.

- Activity recognition on fitness trackers.

PyTorch Mobile

Custom model training and inference.

- Voice analysis for mental health diagnostics.

Research Directions for ML Frameworks

Federated Learning Algorithms

Optimize for TensorFlow Lite in RPM systems.

Patient-Centric Apps

Develop healthcare applications with PyTorch Mobile.

Communication and Networking

1 **Protocols**

MQTT, CoAP, 5G for efficient data exchange

2 **5G Enabled**

Real-time triage and data transfer in ambulances

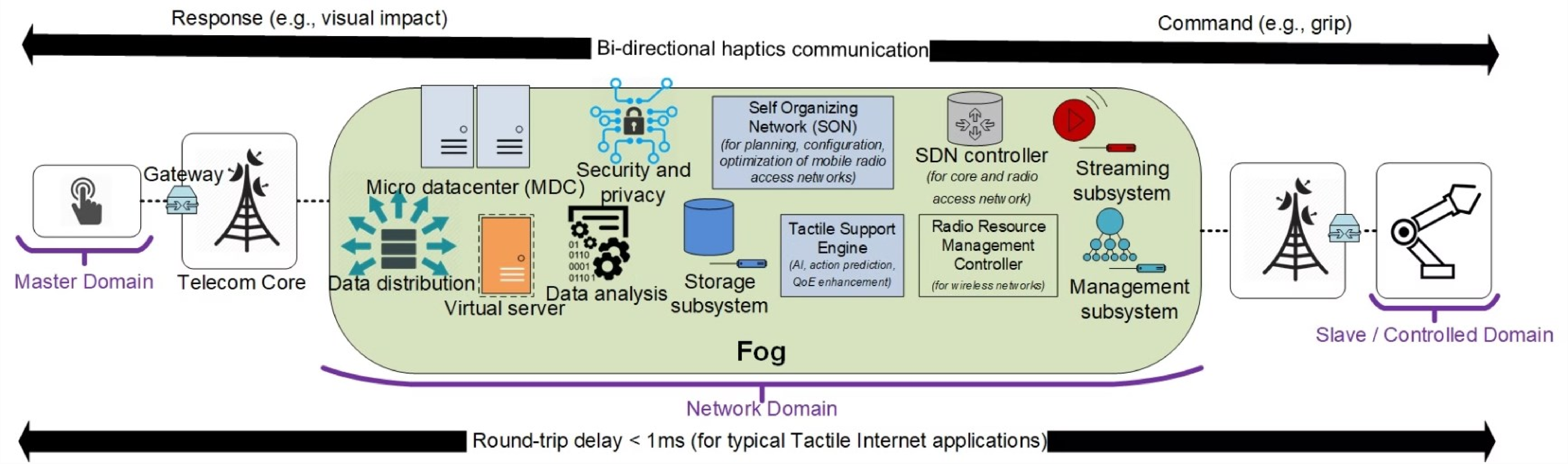
3 **Secure Connections**

Protect patient data and maintain privacy

4 **Low Latency**

Critical for remote monitoring and diagnosis

Edge / Fog Computing in 5G Tactile Internet



Aazam, M., Harras, K. A., & Zeadally, S. (2019). Fog computing for 5G tactile industrial Internet of Things: QoE-aware resource allocation model. IEEE Transactions on Industrial Informatics, 15(5), 3085-3092.

Research Directions for Networking

5G Integration

Enhanced communication for edge systems.

Blockchain Security

Authentication protocols for IoT devices.

Research Challenges & Future Directions

Edge intelligence requires further exploration.

Security, privacy, and ethical considerations.

Key Challenges in Edge Intelligence

Model Optimization

Balance accuracy with latency.

Data Heterogeneity

Standardize across edge devices.

Privacy

Comply with HIPAA and GDPR.

Ethical Considerations

Privacy-preserving ML

Protecting sensitive healthcare data.

- Federated learning
- Homomorphic encryption

Bias Mitigation

Fair and equitable ML models.

Ethical Considerations

Patient Privacy

Safeguarding sensitive medical information.

Transparency and Accountability

Clear understanding of AI decision-making.

Bias Mitigation

Preventing bias in AI models.

Emerging Trends in Healthcare

Neuromorphic Chips

Energy-efficient AI processing for healthcare.

Blockchain for Healthcare

Secure and decentralized data management.

The Future of Smart Healthcare

A world where technology seamlessly integrates with healthcare.

Empowering patients and providers.

Shadow AI in Smart Healthcare

Shadow AI acts as a silent guardian, monitoring and improving existing AI systems in healthcare.

It leverages the power of machine learning to enhance the accuracy, efficiency, and security of traditional AI models.

Summary

Edge Intelligence in Healthcare

The future of healthcare lies in the integration of edge intelligence. It combines edge computing, blockchain, and artificial intelligence to optimize healthcare operations.

Benefits and Applications

Edge intelligence can improve patient outcomes, enhance diagnostics, and ensure data privacy and security.

Research Directions

Further research is needed in model optimization, data heterogeneity, and privacy preservation.

Ethical Considerations

Key concerns include patient privacy, transparency, and bias mitigation.

Thank You!

Q&A

Happy to answer any questions you may have.

Contact:

aazam@ieee.org

maazam@cmu.edu